

Seminar Report

On

AI-Enhanced Remote Classroom

Submitted by

Name -Saurav Kumar Reg No.-12218369

Bachelor of Technology

IN

(Computer Science and Engineering)

Under the Supervision of

Nidhi Mangal

Designation



**LOVELY PROFESSIONAL
UNIVERSITY PUNJAB**

DECLARATION

I hereby declare that the seminar report titled “AI-Enhanced Remote Classroom” submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a record of my own work carried out during the academic session 2026.

I further declare that this report has not been submitted, either in part or in full, to any other institution or university for the award of any degree or diploma.

I confirm that the content of this report is original and prepared by me. Any references used have been duly acknowledged. I also declare that the use of Artificial Intelligence (AI) tools, if any, has been minimal and the AI-generated content in this report is **less than 10%**, ensuring that the majority of the work reflects my own understanding and effort.

I take full responsibility for the authenticity and originality of the work presented in this report.

Name of the Student: Saurav Kumar

Registration Number: 12218369

Course: CSE435

Signature of the Student: saurav kumar

Date: 16/04/26

TABLE OF CONTENTS

Chapter-1: Introduction	4
1.1 Title of the Seminar Topic	4
1.2 Background and Importance of the Topic.....	4
1.3 Objectives of the Seminar	4
1.4 Brief Overview of Approach and Methodology.....	5
Chapter-2: Literature Review	6
2.1 Overview of E-Learning Platforms	6
2.2 Artificial Intelligence in Education	6
2.3 Real-Time Communication in Online Classrooms.....	6
2.4 Role-Based Access Control in Educational Systems.....	7
2.5 Research Gap.....	7
Chapter-3: Conceptual Study / Seminar Work	8
3.1 System Architecture.....	8
3.2 Technology Stack	8
3.3 User Roles and Access Control	8
3.4 Core Modules and API Design.....	9
3.5 AI Recommendation Engine	9
3.6 Real-Time Live Classroom	10
3.7 Low-Bandwidth Optimizations	11
3.8 Security Architecture	11
Chapter-4: Results and Discussion.....	12
4.1 Key Observations from the Study	12
4.2 Comparative Analysis of Existing Platforms	12
4.3 Interpretation of AI Recommendation Output	13
4.4 Advantages of the Platform.....	13
4.5 Limitations and Challenges.....	13
4.6 Insights Gained.....	14
Chapter-5: Conclusion and Future Scope	15
5.1 Summary of Seminar Work	15
5.2 Major Learning Outcomes	15
5.3 Conclusions Drawn from the Study.....	15
5.4 Future Scope.....	16

Chapter-1: Introduction

1.1 Title of the Seminar Topic

AI-Enhanced Remote Classroom Platform for Rural Colleges is the subject of this seminar. This topic focuses on the design and conceptual study of a full-stack web application that integrates Artificial Intelligence, real-time communication, and role-based access to deliver quality education to students enrolled in rural and semi-urban colleges across India.

1.2 Background and Importance of the Topic

India is home to over 39,000 colleges spread across urban and rural regions. A significant proportion of these institutions face acute shortages of qualified subject-matter experts, particularly in emerging technical disciplines. Students in such colleges often lack access to experienced faculty, advanced course material, and the interactive learning environment that their urban counterparts enjoy. This disparity contributes to a widening gap in educational outcomes between urban and rural populations.

The advent of widespread internet connectivity, even in semi-rural areas, has created an opportunity to address this inequity through technology-driven solutions. Remote learning platforms have demonstrated their efficacy on a global scale, especially during the COVID-19 pandemic, when they became the primary mode of education for millions of students. However, most commercially available platforms are designed for high-bandwidth urban environments and lack the personalization, localization, and role-specific features needed for effective deployment in rural India.

The AI-Enhanced Remote Classroom Platform addresses these limitations by combining lightweight frontend design, adaptive AI recommendations, and real-time collaborative tools in a single integrated system. By connecting expert faculty from urban institutions with students in rural colleges, the platform aims to democratize access to quality education.

1.3 Objectives of the Seminar

The primary objectives of this seminar are as follows:

- To study the architecture of a full-stack web application designed for remote education in rural settings.
- To analyze the role of Artificial Intelligence in personalizing the learning experience for individual students.
- To examine how real-time technologies such as Socket.io enable live classroom interactions in low-bandwidth environments.

- To understand role-based access control and its significance in multi-stakeholder educational platforms.
- To identify the advantages, limitations, and future scope of such a platform in the Indian education ecosystem.

1.4 Brief Overview of Approach and Methodology

This seminar is conducted through a conceptual and analytical study of the platform's design documentation, technical specifications, and architectural blueprints. The methodology involves a review of the technology stack comprising React.js 18, Node.js, Express.js, MongoDB, and Socket.io. The AI engine is examined through its rule-based recommendation logic, and the system's security, scalability, and usability are evaluated against established best practices in educational technology.

The study is organized into five chapters. Following this introduction, Chapter 2 reviews existing literature on e-learning platforms and AI in education. Chapter 3 provides a detailed conceptual study of the platform's architecture and features. Chapter 4 presents a discussion of observations, comparisons, and insights. Chapter 5 concludes the report with a summary of findings and proposed future directions.

Chapter-2: Literature Review

2.1 Overview of E-Learning Platforms

The field of e-learning has evolved significantly over the past two decades. Early platforms such as Blackboard (1997) and Moodle (2002) established the foundational concept of Learning Management Systems (LMS). These systems provided tools for content delivery, assignment management, and basic communication between educators and students. However, they were primarily designed for institutions with robust IT infrastructure and did not account for low-bandwidth or resource-constrained environments.

More recent platforms such as Google Classroom, Microsoft Teams for Education, and NPTEL have expanded the scope of digital learning. NPTEL, developed through a joint initiative of Indian Institutes of Technology, has made high-quality course content available free of charge to students across India. However, these platforms largely operate in an asynchronous mode and do not offer real-time interactive classrooms or AI-driven personalization at the individual student level.

2.2 Artificial Intelligence in Education

The application of Artificial Intelligence in education, often referred to as AI-ED, has gained considerable research interest in the last decade. Key areas of AI-ED research include Intelligent Tutoring Systems (ITS), automated assessment, learning analytics, and recommendation systems. Intelligent Tutoring Systems such as Carnegie Learning and ALEKS adapt instructional content based on individual learner performance, providing a personalized learning trajectory that static content platforms cannot offer.

Rule-based recommendation engines represent an accessible entry point into AI-ED without the data requirements of machine learning models. Such systems use predefined logic derived from pedagogical research to classify learners into performance bands and suggest appropriate interventions. Studies have shown that even simple rule-based feedback systems can significantly improve student engagement and performance when integrated into online platforms.

2.3 Real-Time Communication in Online Classrooms

Real-time communication technologies have been critical to replicating the interactive nature of physical classrooms in virtual environments. WebRTC, WebSockets, and libraries such as Socket.io enable bidirectional communication between clients and servers with minimal latency. Research by Bower et al. (2017) demonstrated that synchronous online sessions are more effective for complex conceptual learning compared to purely asynchronous alternatives, provided that network conditions are adequate.

For rural India, network reliability remains a significant constraint. Studies on digital education in Tier-3 cities and villages indicate that 4G connectivity is now available in many rural areas, though bandwidth remains inconsistent. Platforms designed with progressive loading, WebSocket-first transport, and server-side pagination are better suited to these environments than those relying on continuous video streaming.

2.4 Role-Based Access Control in Educational Systems

Multi-stakeholder educational systems require sophisticated access control mechanisms. Role-Based Access Control (RBAC) is a well-established paradigm in enterprise and educational software. Research on RBAC in LMS contexts highlights the importance of clearly differentiated permissions for administrators, instructors, and students to maintain data integrity and privacy. JWT-based authentication combined with server-side role authorization provides a stateless, scalable solution appropriate for web-based educational platforms.

2.5 Research Gap

While significant work has been done on individual components — AI recommendations, real-time classrooms, and LMS design — there is a relative scarcity of integrated platforms that address all three requirements simultaneously while targeting the specific constraints of rural Indian higher education. The platform studied in this seminar represents an attempt to bridge this gap by combining these technologies in a single, cohesive, production-ready application.

Chapter-3: Conceptual Study / Seminar Work

3.1 System Architecture

The AI-Enhanced Remote Classroom Platform follows a client-server architecture with a decoupled frontend and backend. The frontend, built using React.js 18 with Tailwind CSS, communicates with the backend REST API through Axios, a promise-based HTTP client. The backend, powered by Node.js and Express.js, handles business logic, authentication, file management, and real-time events. MongoDB, a NoSQL document database, provides flexible schema management through the Mongoose Object Data Modeling (ODM) library.

The architecture is depicted conceptually in Figure 3.1. The three-tier structure separates the presentation layer (React.js), the application layer (Node.js/Express.js), and the data layer (MongoDB), enabling independent scaling and maintenance of each component.

Figure 3.1: System Architecture of the AI-Enhanced Classroom Platform

3.2 Technology Stack

Table 3.1 summarizes the technology stack employed across the various layers of the platform. Each technology was selected for its open-source availability, active community support, and suitability for the specific requirements of a low-bandwidth rural education environment.

Layer / Component	Technology Used
Frontend	React.js 18 (Hooks) + Tailwind CSS
Backend	Node.js + Express.js
Database	MongoDB (Mongoose ODM)
Authentication	JWT + bcryptjs
Real-time Communication	Socket.io
AI Engine	Rule-based Recommendation System
Data Visualization	Recharts
File Upload	Multer (local) / Cloudinary
Email Service	Nodemailer

Table 3.1: Technology Stack Used in the Platform

3.3 User Roles and Access Control

The platform defines four distinct user roles, each with a specific set of permissions enforced through JWT-based middleware on the backend. The Role-Based Access Control (RBAC) workflow is illustrated in Figure 3.2.

- Administrator: Manages all users, approves faculty accounts, and accesses platform-wide analytics.
- Expert Faculty: Creates and manages courses, uploads study materials, schedules live sessions, and grades assignments.
- Local Faculty: Marks student attendance, monitors engagement within their institution, and facilitates in-class discussions.
- Student: Browses and enrolls in courses, attends live classes, submits assignments, and receives AI-personalized recommendations.

Figure 3.2: Role-Based Access Control Workflow

3.4 Core Modules and API Design

The platform exposes a RESTful API with endpoints organized by functional domain. Table 3.2 presents a representative subset of the available API endpoints and their purposes.

Endpoint	Description
POST /api/auth/register	Register a new user account
POST /api/auth/login	Authenticate user and return JWT token
GET /api/courses	Retrieve paginated list of courses
POST /api/courses	Create a new course (Expert/Admin only)
POST /api/courses/:id/enroll	Enroll student in a course
POST /api/assignments/:id/submit	Submit assignment with file upload
POST /api/attendance	Mark student attendance
GET /api/recommendations/:id	Fetch AI-generated recommendations
POST /api/live-sessions	Create a new live session

Table 3.2: API Endpoint Summary

3.5 AI Recommendation Engine

The recommendation engine, located in the backend controller module, applies a set of predefined rules derived from pedagogical best practices to generate personalized suggestions for each student. The engine evaluates two primary metrics: the student's average score across enrolled courses and their average attendance percentage. Based on

these metrics, it generates recommendations with an associated priority level, a descriptive message, and an action label. The logic is summarized in Table 3.3.

Condition	Priority	Recommendation Action
Score < 40%	High	Suggest beginner materials and remedial sessions
Attendance < 60%	High	Recommend watching recorded lectures
Score 40–70%	Medium	Assign practice problems and intermediate content
Score > 70%	Low	Offer advanced materials and challenge problems

Table 3.3: AI Recommendation Rules

The engine is designed with ML-readiness in mind. The core recommendation generation function can be replaced with a trained machine learning model without modifying the API contract, provided the model returns data in the same structured format: { type, priority, message, action, icon }. Figure 3.3 illustrates the logic flowchart of the current rule-based implementation.

Figure 3.3: AI Recommendation Engine Logic Flowchart

3.6 Real-Time Live Classroom

Live classroom functionality is implemented using Socket.io, which establishes a persistent WebSocket connection between the browser client and the server. When a faculty member starts a live session, a unique room identifier is created. Students joining the session are added to this room via the join-room event. All subsequent interactions — chat messages, hand raises, polls, and announcements — are broadcast within the room. Table 3.4 lists the Socket.io events and their directions.

Event Name	Direction	Description
join-room	Client → Server	Student joins a live session room
chat-message	Client → Server	User sends a chat message
new-message	Server → Client	Message broadcast to all participants
raise-hand	Client → Server	Student signals raised hand
create-poll	Client → Server	Faculty creates an in-class poll
vote-poll	Client → Server	Student submits a vote
poll-updated	Server → Client	Updated vote counts broadcast
user-joined/left	Server → Client	Participant join or exit notification

Table 3.4: Real-time Socket.io Events

Socket.io is configured to use the WebSocket transport protocol exclusively, bypassing HTTP long polling. This reduces latency and server load, which is particularly beneficial in low-bandwidth environments. Figure 3.4 illustrates the event flow during a typical live classroom session.

Figure 3.4: Real-time Socket.io Event Flow

3.7 Low-Bandwidth Optimizations

Recognizing the network constraints prevalent in rural India, the platform incorporates several bandwidth-reduction strategies:

- React lazy loading and code-splitting reduce the initial JavaScript bundle size, enabling faster first-page loads.
- All API responses are paginated with configurable limit and page parameters, preventing the transfer of unnecessary data.
- File upload sizes are capped at 50 MB through Multer middleware, discouraging large uploads that could saturate low-bandwidth connections.
- Gzip compression is applied at the Express.js layer through the compression middleware, reducing response payload sizes by 60–80%.
- WebSocket-only transport in Socket.io eliminates the overhead of HTTP long polling fallback.

3.8 Security Architecture

Security is implemented through multiple complementary mechanisms. JWT tokens with a configurable expiry period authenticate API requests, while bcrypt with a salt factor of 12 secures stored passwords. Helmet.js sets HTTP security headers to protect against common web vulnerabilities. Rate limiting is applied at 100 requests per 15-minute window per IP address, mitigating brute-force attacks. CORS is configured to accept requests only from the designated frontend URL, preventing cross-origin request forgery. Faculty accounts require explicit administrator approval before activation, adding an additional layer of identity verification.

Chapter-4: Results and Discussion

4.1 Key Observations from the Study

The conceptual analysis of the platform yields several significant observations regarding its design philosophy, technical capabilities, and educational applicability:

- The three-tier client-server architecture provides a clean separation of concerns, enabling independent development, testing, and scaling of the frontend, backend, and database layers.
- The rule-based AI recommendation engine, while less sophisticated than machine learning models, is computationally lightweight and requires no training data — a significant advantage in new deployments where historical student data is limited.
- The Socket.io event model supports rich real-time interactions including polls, Q&A, and hand-raising, which are critical for maintaining student engagement in virtual classrooms.
- The four-role RBAC model closely mirrors the stakeholder structure of Indian rural colleges, where distinct responsibilities exist for administrators, visiting expert faculty, local coordinators, and students.
- Low-bandwidth optimizations such as lazy loading, pagination, and gzip compression are thoughtfully implemented, addressing a critical real-world constraint in the target deployment context.

4.2 Comparative Analysis of Existing Platforms

Table 4.1 presents a comparison of the studied platform against commonly used educational platforms on key features relevant to rural education.

Platform	AI Recommendations	Live Polling	Rural Optimized	Bandwidth Need	Target Audience
Google Classroom	No	No	No	Moderate	Global
NPTEL	No	No	No	Low	India
Moodle	Partial	No	No	High	Global
This Platform	Yes	Yes	Yes	Low	Rural India

Table 4.1: Comparison of Existing Platforms

The comparison demonstrates that existing platforms, while feature-rich, do not simultaneously offer AI-driven personalization, real-time interactive tools, and low-

bandwidth optimizations. The studied platform occupies a unique position in this landscape by addressing all three requirements with a focus on rural India.

4.3 Interpretation of AI Recommendation Output

Figure 4.1 and Table 4.1 below illustrate a sample AI recommendation output for a student with mixed performance metrics. The system correctly identifies high-priority intervention needs (low attendance) before addressing medium-priority concerns (moderate scores), demonstrating appropriate priority ordering.

Type	Priority	Message	Action
attendance	High	Attendance in Python is 45%. Watch recorded lectures.	Watch Recorded Lectures
performance	Medium	Score in Data Structures is 55%. Attempt practice assignments.	Practice Assignments
advanced	Low	Score in DBMS is 82%. Explore advanced projects.	Advanced Projects

Figure 4.1: Sample AI Recommendation Output

The recommendation messages are human-readable and directly actionable, which is essential for students who may not have access to academic advisors. The system provides a digital proxy for personalized mentoring.

4.4 Advantages of the Platform

- Democratizes access to expert faculty for students in geographically isolated institutions.
- AI recommendations provide individualized academic guidance at scale without additional human resource costs.
- Open-source technology stack (React.js, Node.js, MongoDB) minimizes licensing costs, making the platform feasible for budget-constrained rural institutions.
- The approval workflow for faculty accounts ensures quality control and prevents unauthorized access to teaching privileges.
- Socket.io-powered live sessions replicate the interactivity of physical classrooms with features such as polls, hand-raising, and real-time Q&A.

4.5 Limitations and Challenges

- The rule-based AI engine does not adapt over time. A machine learning model trained on student performance data would provide more nuanced and accurate recommendations.

- The current implementation does not include video conferencing, which is a significant component of live classroom experiences. Integration with WebRTC or a third-party video API would be necessary for full virtual classroom functionality.
- Local file storage for uploads (via Multer) is not scalable for multi-institution deployment. A cloud storage solution such as Cloudinary or AWS S3 would be required in production.
- The platform does not currently support regional Indian languages, which may pose a barrier for students whose primary language is not English.
- Real-time assessment and proctoring capabilities are absent, limiting the platform's utility for formal examinations.

4.6 Insights Gained

The study reveals that building an effective educational technology platform for rural India is not merely a software engineering challenge, but a socio-technical design problem. Effective solutions must account for the human, institutional, and infrastructural context of deployment alongside technical requirements. The platform's design choices — conservative file size limits, WebSocket-only transport, and a lightweight rule-based AI engine — demonstrate a sensitivity to this context that is often absent in general-purpose educational software.

Chapter-5: Conclusion and Future Scope

5.1 Summary of Seminar Work

This seminar presented a detailed conceptual study of an AI-Enhanced Remote Classroom Platform designed for rural colleges. The platform integrates a React.js frontend, a Node.js/Express.js backend, a MongoDB database, and a Socket.io real-time communication layer into a cohesive full-stack application. The system supports four distinct user roles — Administrator, Expert Faculty, Local Faculty, and Student — each with tailored dashboards and capabilities. A rule-based AI engine provides personalized academic recommendations, and multiple low-bandwidth optimizations ensure usability under the network constraints common in rural India.

5.2 Major Learning Outcomes

Through this seminar, the following key learning outcomes were achieved:

- A thorough understanding of full-stack web application architecture using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Insight into the design and implementation of rule-based AI recommendation systems for educational personalization.
- Knowledge of real-time web communication using Socket.io, including event-driven architecture and room-based broadcasting.
- Understanding of JWT-based authentication and Role-Based Access Control in multi-stakeholder systems.
- Appreciation of the socio-technical dimensions of designing technology for underserved populations.

5.3 Conclusions Drawn from the Study

The following conclusions are drawn from this study:

- Technology alone is insufficient to solve educational inequity; effective platforms must be designed with an acute awareness of the infrastructural, cultural, and economic context of their target users.
- A rule-based AI recommendation engine provides a practical and immediately deployable alternative to machine learning models in environments where training data is scarce.
- The MERN stack offers an excellent foundation for educational web applications due to its flexibility, scalability, and strong community support.

- Real-time interactivity is a critical differentiator for online learning platforms, particularly in contexts where student motivation and engagement are at risk due to the lack of face-to-face interaction.
- Open-source technologies can enable the development of production-grade educational platforms at a fraction of the cost of proprietary alternatives, making them particularly suitable for resource-constrained rural institutions.

5.4 Future Scope

Several enhancements can be pursued to extend the capabilities and impact of this platform:

- **Machine Learning Integration:** The rule-based AI engine can be replaced with a trained ML model — such as a decision tree or collaborative filtering algorithm — that learns from historical student performance data to generate more accurate and adaptive recommendations.
- **Video Conferencing:** Integration with WebRTC or a third-party service such as Daily.co or Agora would enable full video and audio streaming within live sessions, significantly enriching the classroom experience.
- **Regional Language Support:** Adding multi-language support for major Indian languages such as Hindi, Tamil, Telugu, and Marathi would dramatically expand the platform's accessibility for students and faculty who are more comfortable in their native languages.
- **Mobile Application:** Developing companion mobile applications for Android and iOS would cater to the reality that smartphones are often the primary — and sometimes only — internet-connected device available to rural students.
- **Automated Proctoring:** Incorporating computer vision-based proctoring for online examinations would enable the platform to support formal assessment, a capability currently absent.
- **Offline Mode:** A service worker-based offline mode would allow students to access pre-downloaded course materials and complete assignments even without an active internet connection, addressing the challenge of intermittent connectivity in rural areas.
- **Analytics Dashboard:** An enhanced analytics module providing granular insights into learning patterns, engagement metrics, and performance trends would help administrators and faculty make evidence-based pedagogical decisions.

REFERENCES

- [1] Bower, M., Dalgarno, B., Kennedy, G., Lee, M. W., & Kenney, J. (2017). Design and implementation factors in blended synchronous learning environments: Outcomes from a cross-case analysis. *Computers & Education*, 86, 1–17.
- [2] Baker, R. S. J. D. (2016). Stupid tutoring systems, intelligent humans. *International Journal of Artificial Intelligence in Education*, 26(2), 600–614.
- [3] VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist*, 46(4), 197–221.
- [4] Anderson, J. R., Corbett, A. T., Koedinger, K. R., & Pelletier, R. (1995). Cognitive tutors: Lessons learned. *The Journal of the Learning Sciences*, 4(2), 167–207.
- [5] NASSCOM Foundation (2021). Bridging the Digital Divide: Challenges and Opportunities in Rural India. NASSCOM Foundation Report.
- [6] Ministry of Education, Government of India (2020). National Education Policy 2020. New Delhi: Ministry of Education.
- [7] MongoDB Inc. (2023). MongoDB Documentation. Retrieved from <https://www.mongodb.com/docs/>
- [8] Socket.io (2023). Socket.io Documentation. Retrieved from <https://socket.io/docs/v4/>
- [9] React (2023). React Documentation. Retrieved from <https://react.dev/>
- [10] JSON Web Tokens (2023). Introduction to JSON Web Tokens. Retrieved from <https://jwt.io/introduction/>

Github link- <https://github.com/saurav4578/projected>

Linkedin-

Video-https://www.linkedin.com/posts/saurav-kumar-614067247_excited-to-share-my-latest-project-edurav-ugcPost-7455628996710608896-4ZYV?utm_source=share&utm_medium=member_desktop&rcm=ACoAAD0OCLABgoEaRPpT2Xc79LKSOMzwZVxYQrc

Photo-https://www.linkedin.com/posts/saurav-kumar-614067247_mernstack-webdevelopment-fullstackdeveloper-ugcPost-7452990785773551616-XDz1?utm_source=share&utm_medium=member_desktop&rcm=ACoAAD0OCLABgoEaRPpT2Xc79LKSOMzwZVxYQrc